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BETWEEN SESSION RELIABILITY OF TRADITIONAL AND TEMPORAL BILATERAL AND UNILATERAL DYNAMIC STRENGTH INDEX CALCULATIONS AND ASSOCIATION WITH SPRINT AND CHANGE OF DIRECTION PERFORMANCE

ABSTRACT

The present study aimed to; 1) evaluate the between session reliability of traditional and temporal based bilateral and unilateral dynamic strength index (DSI); 2) determine the association between DSI and sprint and change of direction performance in participants clustered by DSI root metrics. Thirty-eight recreationally active males (age: 23.4 ± 3.4 years) completed 20-m sprints, 5-0-5 change of direction test, bilateral and unilateral countermovement jumps (CMJ), and isometric mid-thigh pulls (IMTP) on two occasions. Vertical ground reaction force profiles were assessed to determine traditional DSI (DSI_T), DSI based on mean force achieved after 100ms (DSI_{100}), 150ms (DSI_{150}), the entire CMJ propulsive phase (DSI_P), and CMJ propulsive impulse (DSI_I). Bayesian interclass correlation coefficients and multiple linear regression following K-means clustering were used to analyze the data. Between session reliability of DSI was poor-moderate (ICC = 0.24-0.76) and highest for DSI_T measures, albeit with some uncertainty (95% HDI: 0.53-0.87). Bilateral temporal based DSI reliability was greater compared to unliteral equivalents, and temporal based DSI calculated for the non-dominant limb, and those determined over a small-time epoch for the dominant limb were poor. There was limited association across DSI measures and straight-line 20m sprint and 5-0-5 change of direction performance test. DSI_{100} for the dominant side had the strongest association with the performance outcomes (R^2 = 0.33-0.36) and was improved in participants that expressed higher CMJ peak propulsive force. DSI should be used with caution with respect to exercise prescription for the intention of improving tasks requiring rapid horizontal center of mass translation.

Key Words: Strength Profiling, Resistance Training, Isometric Midthigh Pull, Vertical Jump, Force Platform

INTRODUCTION

Given the importance to athletic performance and sporting success, strength and power profiling is essential to athlete monitoring (32). Within this context, force platforms have become important to data driven athlete profiling, monitoring and exercise prescription, with recent commercialization and improved useability resulting in increased adoption in applied sport and exercise settings. Given their reliability and association with sport specific strength and power activities (9,22,34,35), force-time analysis of vertical jumps and isometric strength profiling of the lower limbs have become commonplace, where the wealth of data challenges the user in determining meaningful outcome variables. This challenge is aggravated by an increase in 'novel' force-time metrics that are often adopted without thorough supporting evidence. One such metric is the dynamic strength index (DSI), where it is suggested that the ratio between peak force achieved in a vertical jump and peak force in an isometric mid-thigh pull (IMTP) or squat provide insight regarding an athlete's strength or ballistic training need (44). Despite this, evidence assessing the value and utilization of the DSI for exercise prescription is lacking and further compounded by recent studies proposing more efficacious methodological and analytical approaches to determining the DSI.

Specifically, the DSI evaluates a participant's ability to express force during a ballistic task relative to their maximal force producing capacity, a foundational characteristic important for movement effectiveness across various sporting tasks. A higher DSI (>0.8) denotes high force expression, indicating a strength training need, whilst a low value (<0.6) denotes a reduced ability to express maximal force during a ballistic task, thus indicating a ballistic training need (44). While the theoretical constructs of the DSI have been considered, it is essential to evaluate its reliability and validity to determine its usability in exercise prescription. In the most part, current evidence is focused on determining the between session reliability of DSI (12,44,49). Previous work has demonstrated excellent reliability (ICC =.97) of DSI determined using a squat jump (SJ) and IMTP in a population of male college level athletes (49). However, there is the suggestion that in youth athletes DSI determined using a CMJ offers superior between session reliability attributed to reduced repeatability of the SJ (12). More recently, the biomechanical principles underpinning the value of the traditional DSI, determined from peak force in bilateral strength and power assessments, have been questioned (4). This has led to the suggestion that unilateral DSI evaluation and consideration of the time needed to generate force (impulse) in each test provide better insight into the mechanical constraints that limit lower limb power production and in turn may improve the application of DSI for exercise prescription. Bishop, et al. (2) suggested that single leg DSI may better reflect the unilateral demands of team sport specific tasks such as sprinting, change of direction (CoD) and jumping, and demonstrated moderate to good reliability (ICC =

0.71–0.79) of single leg DSI in recreational sport athletes when determined using an isometric squat and a CMJ. Recently, temporal based DSI calculations have been suggested to be more appropriate given that peak force used in the traditional DSI has poor association to dynamic tasks, peak force in the IMTP occurs later than in a vertical jump, the importance of impulse in determining the outcome of dynamic tasks, and considering that decisive actions in sport occur in a much quicker timeframe than when IMTP peak force is achieved (24,28). Haischer, et al. (24) suggested that impulse based DSI may be more appropriate and demonstrated this to some extent to be distinct from traditional DSI. James and Comfort (28) later demonstrated acceptable absolute reliability of bilateral DSI calculated as mean force over ecologically relevant time epochs and that such measures were distinct. Whilst there is some support for reliability of DSI across methods of calculation, the reliability of temporal based DSI calculations in unilateral assessments is yet to be determined.

Sheppard, et al. (44) first defined DSI thresholds for determining strength, ballistic or concurrent training needs, and although the values advocated are arbitrary and not based on empirical evidence, they have been heavily adopted. Importantly, the proposed thresholds were based on a case study approach of five individuals (some recovering from injury) where bespoke resistance training was effective in reducing DSI. Furthermore, DSI has been shown to reduce following strength training (13) which would appear to support the primary context of this measure. However, only one study has evaluated the efficacy of traditional DSI when resistance training is prescribed based on a DSI pre-screen. In this study, a five-week in-season resistance training program guided by DSI was found to be effective in improving sprint and change of direction performance in a small sample of high-level basketball players, compared to a decline in performance observed in a control group receiving “regular resistance training” (40). Whilst this study provides valuable insight, the limited evidence base, and a lack of consideration of temporal based DSI calculations highlights the need for further research. Specifically, robust assessment in this context requires participants to be matched based on baseline vertical jump and isometric strength performance. Training should then be prescribed both in alignment with and in contrast to the recommendations derived from the DSI. This approach is yet to be adopted and presents a considerable logistical challenge. Considering this, and the current lack of evidence based DSI thresholds to inform training interventions, a valuable method for evaluating the utility of the DSI in exercise prescription is to investigate its association with key performance measures. For many practitioners, this represents the potential of an exercise intervention that is guided by prior data to elect improvement in sport-specific performance outcomes. Based on the theoretical foundations of the DSI, among athletes with comparable muscular strength capacities, those who can express force more effectively during dynamic tasks should exhibit superior performance in

activities where muscular power is a limiting factor. Therefore, a strong association between DSI and measures of athletic performance would provide meaningful evidence to support the integration of DSI in resistance exercise programming and provide context for the value of this measure for athlete monitoring.

Previous studies specific to traditional DSI have demonstrated differences based on sex and sport (48), and association with IMTP and CMJ force-time metrics (42), lower body muscle tendon complex stiffness (42), but not CMJ and SJ height (43). Beyond this, there is little consideration of the association between DSI and activities such as sprinting and CoD that are important to athletic performance and often the focus of resistance training interventions for team sport athletes. Recent work by Pleša, et al. (38), demonstrated that DSI measured in trained male basketball players was weakly associated with 5-0-5 CoD performance but not straight-line sprint speed. Curiously, this association was only apparent in season but not during the pre-season. Further analysis by the same group reported weak correlations between DSI and unilateral horizontal jump ($r = -0.36$; $p < 0.05$), lateral jumps, linear sprint and CoD deficit when assessed pre-season (39). However, where there have been attempts to evaluate the link between DSI and other performance outcomes, previous work fails to control for the metrics used in determining DSI where it is possible to have a low DSI indicating a ballistic training need, when in fact both strength and the ballistic expression of that strength are low. In this case, strength training would be beneficial, contradictory to DSI convention (46). Controlling for the metrics used to determine DSI, the present study will for the first time examine the association between DSI and sprint and CoD performance. Furthermore, the study will evaluate if the strength of the associations differs between the bilateral and unilateral DSI and temporal DSI calculations. Such work is essential to aid in the understanding of the useability of the DSI and to refine the most appropriate methods of calculation.

Specifically, the study aims were twofold; 1) to evaluate the between session reliability of traditional and temporal based calculations of bilateral and unilateral DSI; 2) to determine the association between DSI calculations and sprint and CoD performance in participants clustered by DSI root metrics.

METHODS

Experimental Approach to the Problem

A cross-sectional study with a test-retest design was used to determine the between session reliability of traditional and temporal based DSI calculations determined from bilateral

and dominant and non-dominant unilateral assessments of CMJ and IMTP. Participants were then grouped by performance in the CMJ and IMTP and the association between traditional and temporal based DSI and straight-line sprint speed and CoD performance were evaluated.

Subjects

Following ethics approval from the host institute (P146580) and informed written consent, thirty-eight recreationally active males (Mean±SD; age = 23.4 ± 3.4 yrs; height = 180.1±6.3 cm; body mass = 80.5±10.0 kg) participated in the study. Participants were university students who played recreational or university level sports (including soccer, hockey, athletics, cycling and combat sports) and were free from musculoskeletal injury or health contradictions that prevented safe completion of the physical tasks confirmed via the completion of a health screen survey. Participants were recruited via opportunistic sampling via word of mouth and were not specifically familiar with the physical assessments employed.

Procedures

Participants were asked to attend the human performance laboratory at the host institute on three separate occasions. In the first session, shoes and heavy clothing were removed and body mass (Digital Scales; Model 707, SECA Instruments Ltd., Germany) and stature (Stadiometer, SECA Instruments Ltd., Germany) assessed to the nearest 0.1kg and 0.1cm respectively. Participants were then familiarized to all testing procedures before completing two experimental trials. Each visit was separated by at least seven days and performed at the same time of day to control for circadian variation. Participants were asked to abstain from intense physical activity in the 48 hours prior to each visit. On each occasion participants completed the assessments outlined below following a standardized twenty-minute dynamic RAMP protocol warm-up (29), designed to specifically target the demands of forthcoming exercises. Participants completed 5 minutes of stationary cycling at a fixed load of 70 watts, followed by 5 minutes of self-selected dynamic stretching, and 10 body weight squats and 10 sub maximal CMJs. Prior to the sprinting and change of direction tasks, participants completed attempts of each task at 50%, 70% and 90% of their perceived maximal effort.

Countermovement Jump

Participants completed three bilateral CMJs whilst standing on two uni-axial force platforms (PASCO, California, USA) each sampling at 1000Hz. Participants were asked to stand as still as possible for approximately 5 s to obtain a stable force trace before executing the jump. Participants were asked to jump "*as high and as fast as possible*" with arms akimbo (fixed on the hips), following a three second countdown (36). Participants were informed that

their legs should remain straight during flight, contacting the ground with the tips of their feet and with knees extended (36). Correct technique was ensured prior to completion of the recorded trials. Attempts that deviated from this pattern were discarded. Each attempt was separated by 60 s of rest with the attempt that elicited the greatest jump height used in further analysis. Unilateral CMJs were assessed in the same manner for both the dominant and non-dominant limbs, where for the non-standing limb participants were asked to flex the knee to approximately 90°. Limb dominance was based on participant preference.

Isometric Mid-Thigh Pull

IMTP trials were carried out as per the recommendations by Comfort, et al. (9) using a purpose built rack (Indigo Fitness, Warwickshire, UK). For each participant, the bar height was adjusted to achieve a knee angle of 125-145° and a hip angle of 140-150°, measured using a goniometer. Participants stood on two uni-axial force platforms (PASCO, California, USA) each sampling at 1000Hz, with feet hip width apart. Body position was further adjusted so that the participants had an upright torso, shoulders were placed above the bar and ankles dorsiflexed. To ensure maximal force application, lifting straps were used to secure the hands to the bar. Participants were instructed to stand as still as possible for a period of 1 s applying a standardized pretension force of ~100N. A 55" television (MultiSync Me551, NEC, Tokyo) was positioned ~3m in front of the participant for visual feedback. Participants were instructed to "*push into the ground as hard and as fast as possible*" following a three second countdown and strong verbal encouragement was provided so that the participant maintained a maximal effort for ~5s. Attempts were discarded if there was an absence of a stable pretension period, a countermovement at initiation, or if the peak force occurred at the end of the trial. Participants completed two attempts with 2 minutes rest between trials. Distance between the hands, heels and hallux were measured and used to aid positioning in subsequent trials. Unilateral and IMTP performance for the dominant and non-dominant leg were assessed in the manner previously outlined with a single force plate positioned centrally to the midpoint of the participant's grip. In each case, the IMTP that elicited the greatest peak force was used for further analysis.

Sprint and Change of Direction Ability

From a standing start, participants completed assessment of 20m sprint speed on an outdoor athletics track. Timing gates (Brower T-C, Utah), mounted on tripods positioned at approximately hip height, were placed at the start and at 5-, 10- and 20-m. Participants were positioned 0.5m behind the start line with head/chest 5-10cm away from the first timing gate (26). Participants were instructed that the first movement should be forward and to sprint beyond the final timing gate. Sprint time was recorded to the nearest 0.01s. Participants

completed three attempts separated by 60 s of rest and the attempt that elicited the fastest time was used for further analysis.

Participants then completed the 5-0-5 CoD run (18). From a standing position, participants were instructed to sprint as quickly as possible, immediately pivot from their dominant leg at a point marked at 15m, then sprint as fast as possible in the opposing direction. A timing gate mounted on a tripod at approximately hip height was placed at 10m and was triggered to start at the first pass through and then triggered to end timing at the second pass. Time was recorded to the nearest 0.01s. Participants completed three attempts separated by 60 s of rest and the attempt that elicited the fasted time was used for further analysis.

Data Analysis

Analysis of CMJ and IMTP force-time-curves was completed using a custom written Microsoft Excel (v2016, Microsoft Corporation, Washington, USA) sheet. Raw unfiltered vertical force from each platform was exported and summed for evaluation of bilateral CMJs and IMTPs. The use of raw unfiltered force data for this purpose has been advocated (9,16,25) and represents the approach typically adopted for these assessments (10).

For the CMJ, as per the procedures outlined in previous work (6), force-time data was mathematically integrated to determine velocity at each sampling point. Body weight (N) was determined in the 1 s quiet standing period prior to the start of the movement. The propulsive phase was determined as the period from when velocity first exceeded zero following initiation of the movement to the point of take-off. Within this period, peak net force, mean net force, mean net force in the initial 100ms and 150ms, and propulsive net impulse were determined using the trapezoid method (Figure 1.).

For the IMTP, in accordance with published procedures (9), the onset of the movement was determined at the point where force exceeded 5*SD of the standing noise determined in the 1s period prior to the test. Peak net force, mean net force in the initial 100ms and 150ms, and impulse and mean force in a time frame that matched the propulsive phase in the CMJ were determined (Figure 1.). Impulse was calculated in the same manner as previously described.

***** Insert Figure 1 Here *****

As suggested in previous work (24,28), traditional DSI (DSI_T) was calculated as peak force in the CMJ divided by peak force in the IMTP. DSI were also calculated in the same way using mean force achieved after 100ms (DSI_{100}), mean force achieved after 150ms (DSI_{150}), mean force achieved in the CMJ propulsive phase (DSI_P) and impulse in the CMJ propulsive phase (DSI_i)

Statistical Analysis

In an attempt to avoid misinterpretation of commonly reported p values (7), Bayesian statistical analysis was used. Bayesian analysis allows direct probability statements, expresses the uncertainty around the parameter values, and is more appropriate for studies with smaller sample sizes when compared to traditional statistical approaches that utilize p values and confidence intervals (33). Initially, the reliability of IMTP and CMJ data at the previously defined aspects of the force-time curve, including the calculated DSIs, were determined. Thereafter, participants were grouped separately based on IMTP and CMJ peak force for bilateral, dominant and non-dominant limbs using *k*-means cluster analysis (27). Association between the DSI measures and the sprint and CoD outcomes were then evaluated via a series of Bayesian multiple linear regressions. To establish whether DSI values were related to each other, a series of Bayesian linear regressions to establish if these data were independent of each other.

The interclass correlation coefficient (ICC) and 95% limits of agreement (LoA) were used to determine the reliability of force-time, DSI metrics, sprint and CoD time. The ICCs were run using simple linear regression models from data taken from two visits. Each model was then extracted to determine the proportion of variance explained by the participant-level random effects relative to the total variance to calculate the ICC. The ICCs were interpreted as poor (<0.5), moderate (<0.75), good (<0.9) or excellent (>0.9) (30) based on the point estimate of the model, with inclusion of Higher Density Intervals (HDI) to express uncertainty. Although previous work has suggested that the lower bound of a 95% confidence interval of an ICC should be used alongside the point estimate to provide the 'worst case scenario' (30), this study adopted to use the point estimate in conjunction with lower and upper 95% HDI since Bayesian analysis allows for direct probability statements about parameters, making interpretation more intuitive. For example, Bayesian analysis calculates the probability that a parameter falls within a specified range, defined by the lower and upper bounds of the 95% HDI. The point estimate (mean) represents the most likely outcome, with a 95% chance that the true value lies within this interval. The 95% LoA were also conducted in accordance with previous suggestions (21). This approach reports the mean bias (the difference between the two visits), including 95% higher density intervals expressing the upper and lower bounds of agreement, and interpreted as a 95% probability that a future difference lies within these bounds.

Participants were then grouped separately based on IMTP and CMJ peak force for bilateral, dominant and non-dominant limbs using *K*-means cluster analysis. *K*-means clustering was implemented using the facto extra package in R. *K*-means clustering is an iterative process

whereby the distances between input data and several randomly allocated seeds (k) in the data space are determined. This results in k temporary clusters of data each allocated to each seed. For the newly formed temporary clusters the mean of the data contained within the cluster is calculated, thus forming a new centroid; the distance of the data points in the data space to the new temporary centroids is then re-calculated. This process of re-defining the cluster centroids is repeated until the change in positions of the cluster centroids converges on a constant location (19). The number of cluster groups, number of observations, *k*-means cluster centers are summarized in table 1.

***** Insert Table 1 Here *****

A series of Bayesian multiple linear regressions with fixed intercepts and random slopes for the groups (based on higher or lower IMTP or CMJ peak force) were then conducted to establish whether DSI_T, DSI_{100}, DSI_{150}, DSI_P and DSI_I were related to 5-0-5 CoD and 5,10, 20m sprint times. Random slopes were included in the models to extract group-specific slopes, enabling the assessment of how each group's slope varied with the DSI metric. These models were either fitted with gaussian or skew-normal distributions and were then subsequently compared using Leave-One-Out-Cross-Validation to determine the best model for further analysis, with skew distributions demonstrating a better fit. These models also assessed if the grouping of those with higher or lower IMTP and CMJ peak forces have a reduced 5-0-5 CoD and sprint performance. The slopes of each model were then extracted for further analysis where the probability of a difference between the slopes of the higher group and lower group were compared. A slope that is increasingly negative was interpreted as superior as the DSI metric would increase as time to complete the performance task decreased (higher DSI meaning shorter time to complete task). Moreover, to determine the relationship between all the possible DSI calculations for both bilateral, dominant and non-dominant data, a series of simple Bayesian linear regressions were conducted to establish whether if the different methods used to quantify DSI are independent of each other. The R^2 value was used to establish the strength of this association for all regression models conducted in this study, alongside 95% higher density intervals to express uncertainty around these parameter estimates.

For all Bayesian analysis conducted in the study, the priors for the intercept (α) was a student-t distribution with 3 degrees of freedom, the location parameter (μ) was set to the median of the response variable, and the scale parameter (σ) was set to the median absolute deviation of the response variable. The prior distribution for the regression coefficients (β) was a wide normal distribution. All analyses were conducted using R (R Core Team, 2024) using Bayesian Regression Models in Stan (brms) package (5) to implement a Hamiltonian Markov Chain

Monte Carlo with a No-U-Turn Sampler. All models were checked for convergence (R-hat = 1), with graphical prior and posterior predictive checks showing how the predicted distribution compared to the observed data (20). To describe the uncertainty around the estimation reliability and association with performance, lower and upper 95% HDI are reported along with R^2 values, means ± standard deviation.

Results

Between Session Reliability

Between session reliability of all outcome measures is presented in Tables 2-5 and are interpreted based on the point estimate in addition to the 95% HDI. All bilateral CMJ force-time variables showed moderate to good reliability (Table 2, ICC = 0.64-0.80); however, the reliability of propulsive net force ranged from poor to good (Table 2, ICC = 0.64; 95% HDI: 0.41 to 0.81). Unilateral CMJ force-time variables demonstrated poor to good reliability and showed more variation than bilateral CMJs and a broader range of HDI (Table 2, ICC = 0.50-0.72) Propulsive net impulse for the dominant leg showed the greatest between session variation (Table 2. ICC = 0.30; 95% HDI 0.01 to 0.58).

***** Insert Table 2 Here *****

Between session reliability of the IMTP force metrics ranged from poor to good (Table 3. ICC = 0.29-0.86). Peak net force showed almost good reliability in all cases, albeit with some uncertainty (Table 3. ICC > 0.80; 95% HDI (range from lowest to highest: 0.64 to 0.93)). Reliability of the temporal force outcomes was higher in the bilateral IMTP (Table 3. ICC = 0.40-0.78) compared to the unilateral attempts (Table 3. ICC = 0.29-0.47). 100ms mean net force demonstrated the lowest repeatability compared to other outcomes assessed for each test (Table 3. ICC = 0.29-0.48; 95% HDI (range from lowest to highest: 0.01 to 0.70).

***** Insert Table 3 Here *****

Reliability of DSI calculations ranged from poor to moderate (Table 4. ICC = 0.24-0.73). Across all tests, DSI_T showed the greatest reliability (Table 4. ICC= 0.72-0.76), albeit with some uncertainty (95%HDI: (range from lowest to highest: 0.53 to 0.87). Temporal based DSI for bilateral assessments, and DSI_P and DSI_I for the dominant limb, showed moderate reliability (Table 4. ICC = 0.50-0.67), albeit with some uncertainty (95% HDI: (range from highest to lowest: 0.22 to 0.82). The reliability of all other DSI measurers were poor.

***** Insert Table 4 Here *****

Reliability of 20m sprint and 5-0-5 CoD time were good (Table 5. ICC = 0.81 (95% HDI: 0.66 to 0.90) and 0.75 (95% HDI 0.56 to 0.87) respectively), albeit with some uncertainty. The 5m and 10m sprint times demonstrated moderate reliability (Table 5. ICC = 0.61 (95% HDI: 0.36 to 0.80) and 0.69 (95% HDI: 0.44 to 0.84) respectively), albeit with some uncertainty.

***** Insert Table 5 Here *****

Association Between DSI Calculations

Supplementary Table 1 shows the results of the association between the different DSI calculations. There was limited association between the DSI measures across the different tasks (R^2 <.414). Between the different tasks, non-dominant and dominant had the strongest association (R^2=0.697). Within each assessment, DSI_T showed the lowest association with temporal based DSI measures (R^2<.409) and each of the temporal based DSI calculations were more strongly associated (R^2=.616-.934).

Association Between DSI and Sprint & CoD Performance.

Supplementary Table 1 shows the results of the association between the different DSI calculations. There was limited association between the DSI measures across the different tasks (R2 <.62). Within each assessment, DSIT showed the lowest association with temporal based DSI measures (R2<.41) and each of the temporal based DSI calculations were more strongly associated (R2=.57-.94).

***** Insert Table 6 Here *****

Both dominant and non-dominant DSI_T explained between 19.8-25.5% of the variation in the selected performance outcomes (Table 7.). Non-dominant DSI_{100}, explained 13.4-17.3% of the variation in sprint performance times, and all other DSI calculations performed poorly. The probability of a difference between the slopes of the high and low group was less than 67% in all cases.

***** Insert Table 7 Here *****

When clustered by CMJ peak net force, bilateral DSI calculations explained less than 12% of the variation in the selected performance outcomes (Table 8. R^2<.12 in all cases). Only the DSI_{100} model for all sprint distances, and DSI_T and DSI_{150} for 20m sprint time, achieved an R^2 greater than .01. The probability of a difference between the slopes of the high and low group was less than 68% in all cases.

***** Insert Table 8 Here *****

Dominant DSI_{100} explained 33.1-36.3% of the variation in each of the measured performance outcomes (Table 9.). For these analyses, the probability of differences between slopes was 70-82%, indicative of an increasingly negative slope in the high group which infers that a higher DSI resulted in a reduced performance time. The remaining DSI measures explained 10-26% of the variation in each of the measured performance outcomes. However, except for DSI_{150} modelled for 5-0-5 and 10m sprint time, the probability of a difference between the slopes of the high and low group were less than 68% in all cases.

409 Comparatively, non-dominant DSI_T and DSI_{100} explained 12.6%-16.0% of the variation in the
410 measured performance outcomes and were greater than all other non-dominant DSI
411 measures (Table 9. R^2<.99 in all cases). For all DSI_T and DSI_I models, and DSI_P except for
412 20m sprint time, the probability of differences between slopes was 70-74%, indicative of an
413 increasingly negative slope in the high group.

414 ***** Insert Table 9 Here *****

415

DISCUSSION

The present study addressed key knowledge gaps that prevent appropriate evidence-based application of the DSI by examining the between session reliability of both bilateral and unilateral, traditional and temporal based DSI, and provided unique insight into the practical significance of these measures by determining their association with sprint and CoD performance. Results demonstrated that DSI calculations across the different tests were distinct, indicating characterization of different lower limb force producing constructs. Between session reliability ranged from poor to moderate (ICC range = 0.24-0.76) and was highest for DSI_T measures, although 95% HDIs ranged between moderate and good. Bilateral temporal based DSI reliability was greater compared to unliteral equivalents, and temporal based DSI calculated for the non-dominant limb, and those determined over a small-time epoch for the dominant leg were poor. For the most part, there was limited association between DSI measures and straight-line 20m sprint and 5-0-5 COD performance, which typically were not influenced when participants were stratified by DSI input metrics. DSI_{100} for the dominant side had the strongest association with the performance outcomes (R^2 = 0.33-0.36) and performed better in participants that expressed higher CMJ peak propulsive force. Whilst these data are valuable in allowing practitioners to select reliable DSI methods, the application of DSI for profiling, monitoring and exercise prescription in the context of improving rapid horizontal translation of the center of mass in team sport specific athletic performance requires caution.

Data in the present study demonstrate that between session reliability of the DSI is calculation specific. Reliability was greatest for DSI_T across the bilateral and unilateral measures but lower than that reported in previous work (2,12,13,28,44,49), and with some uncertainty as indicated by 95% HDI. Only two studies have considered the between session reliability of temporal based bilateral DSI and have demonstrated, or infer, moderate-good reliability based on calculations using the same force-time characteristics as those in the present work (24,28). Whilst the results of the present study support the suggestion that temporal based DSI calculated across bilateral tests have reduced reliability, although marginally lower, this finding is in keeping with previous work, which similarly demonstrates confidence intervals between poor and good (28). This study is the first to examine the reliability of temporal based DSI on unilateral assessments, which is important given the proposed increased biomechanical relevance. In the most part, the between session reliability of these measures was poor. Collectively, whilst the between session reliability results presented here are acceptable in some cases, they are poorer than that shown in previous work (2,12,13,28,44,49) which may be attributable to the interaction of several minor differences between the present and previous work that are explained herein.

Firstly, although not overtly presented, several studies that have evaluated between session reliability appear to have determined DSI using absolute force data (2,12,13,28,44,49). However, in the present study net force was used and is theoretically more appropriate given that this represents the force generated to execute the movement and represents the force producing capacity of the participant in the selected task. Determining DSI based on net force is therefore essential when evaluating the association with other performance metrics. Using absolute force will distort the calculated DSI given that the force of body weight in the vertical jump assessment (numerator in the DSI calculation) represents a greater proportion of the force achieved than in the isometric test (denominator in the DSI calculation). Removing the force of body weight from the vertical jump and isometric test results in a reduced trial mean DSI and increased standard deviation, meaning between session reliability of DSI determined using absolute force is likely to be superior to that when calculated using net force. Although this effect may be marginal, the impact will be exacerbated by increased between session variation in the DSI input metrics.

Inherently we should expect the between session reliability of ratios to be heavily influenced by the between session reliability of the input metrics. Except for peak net force, the reliability of IMTP temporal force-time characteristics was generally poor compared to outcomes determined from the CMJ. Whilst it is typically considered IMTP force-time metrics are highly reliable, in the most part the literature is specific to measures of peak force (22). Several approaches have been employed to determine rate of force application in the IMTP inclusive of the calculation of rate of force development, impulse and force at specific time epochs (9). Previous work examining the between session reliability of force at specific time epochs is surprisingly limited and somewhat mixed. Although some previous work has demonstrated a high degree of reliability (10,15,17,23), this is not always the case, particularly in short time windows close to the initiation of the pull and under inspection of ICC 95% confidence intervals (1,14,15,17). Poor between session reliability of the IMTP temporal force-time characteristics compared to previous work may be explained by several factors.

Similarly, but distinct from the previous discussion, the reduced reliability of IMTP temporal force-time characteristics may in part be attributable to the calculation of net force. It is common for reliability of IMTP force-time metrics to be determined based on absolute force (17,23,44), however, between session reliability of net force is anticipated to be lower than that calculated using absolute force given a larger relative difference between scores when the force of body weight is subtracted. Furthermore, IMTP reliability may have been influenced by our approach to controlling pre-tension prior to test initiation. Whilst it is recommended that pre-tension be below <50N (9), in almost all studies examining the reliability of IMTP force-time metrics, pre-tension is either not acknowledge or no robust methods is provided to

outlined how pre-tension is controlled. Given the potential for significant impact on temporal based IMTP force-time characteristics, pre-tension was tightly controlled and visual and oral feedback provided to participants which appears unique based on the methodological approaches reported in previous work (1,14,17,23,45). Whilst this is a strength of the methodological approach, it may have influenced reliability of the data collected. Firstly, the effect of pre-tension on IMTP performance is not known and the rigorous control adopted in the present study may be theoretically sound but result in a suboptimal approach. Indeed, it is suggested that adopting an internal focus of attention can be detrimental to force production tasks (31). Visual feedback of the force-time trace in synergy with assessor feedback regarding the control of pretension may have directed attention internally to the movements being executed, which may ultimately influence between session reliability.

Furthermore, although the familiarization approach used in the present study was consistent with previous research examining the reliability of IMTP force–time characteristics (11,14), the participants involved were not specifically resistance-trained. This may have influenced the reliability of the results. However, this effect is unlikely to have been consistent across all participants, as indicated by the broad range of 95% HDIs.

Despite theoretical bases for the application of DSI as an exercise prescription tool, direct evidence pertaining to the ecological validity of DSI is lacking. The present study uniquely examined the association of both traditional and temporal DSI and sprint and CoD performance, where it is considered that a strong association would provide improved evidence-based justification for application. The best performance in each of the assessments was used to aid in mitigating the influence of poor between session reliability demonstrated in a number of measures. Importantly, our results indicate that in the most part, the distinct DSI outcomes shared little relation to each other, indicating the evaluation of distinct force producing capabilities and justifying their inclusion in subsequent regression analysis. However, in all cases the association between DSI calculations and sprint and CoD performance was weak. These findings are in contrast to previous work demonstrating a relationship between DSI_T and specific CMJ and IMTP force-time outcomes (43,47), lower body muscle tendon complex stiffness (42), and the force-velocity profile determined from loaded vertical jumps (37). Such discrepancies are likely attributable to differences in tasks, where the stronger association seen in previous work are specific to outcomes derived from tasks requiring vertical force production or metrics that are included in the DSI calculation. Sprint and CoD performance are substantially influenced by an athlete's ability to generate horizontal force (8,41) and therefore a DSI based on vertical force production may not adequately reflect the demands of the activity.

A particular strength of the approach adopted in the present study was to further consider if the strength of the associations were affected when participants were grouped by DSI input metrics. This is important given that DSI is independent of high or low performance in determining assessments and therefore exploring the association with DSI, and the selected performance outcomes in participants that are similar in their strength and power producing capabilities is a more appropriate and robust approach. Interestingly, albeit still low, in a number of instances the strength of the associations were improved when participants were stratified by CMJ net force. Specifically, DSI determined from the dominant limb consistently demonstrated the highest association with the performance outcomes (R^2=0.105-0.363). Of these measures dominant DSI_{100} had the strongest association (R^2=0.331-0.363), and despite some uncertainty, the results infer that the strength of the association was greater in participants grouped by higher CMJ peak net force. This is particularly intriguing given that based on the approach to analysis, it may be anticipated that the strength of the association would be similar between groups. Even when performance in one of the DSI constructs is poor, in a reasonably homogeneous group it would seem sensible to anticipate that those that can express a greater quantity of their force producing capacity in a dynamic task (i.e. a higher DSI), would show the same trend in association to the selected performance metrics. The fact that this is not the case likely indicates that those with a higher DSI in the low CMJ net force group had a poor strength expression, where in such cases the DSI is less valuable given that the ability of the musculature to generate high force underpins power production (46). Furthermore, the specificity of this finding to the dominant limb may underpin the relevance of single limb assessment over a bilateral test with respect to the performance measures (2), where associations being less prevalent in the non-dominant limb is likely explained by poor familiarity with the movement as indicated by the poorer reliability. Despite these findings, across all the DSI outcomes associations with sprint and CoD performance were minimal, indicating a need for caution in the application of DSI in this context.

Whilst the present work provides important new insight into the utility of the DSI, it is not without limitation. Previous work has examined the DSI using a combination of either the squat jump (SJ), CMJ, IMTP and isometric squat (2,12,44). It should therefore be noted that application of the results of the present study are specific to the DSI calculated using a CMJ and an IMTP. The isometric squat has been shown to elicit greater peak force than the IMTP (3), with greater force attributed to the action of the back extensors. Thus, the DSI calculated using an isometric squat is likely to be reduced compared to that from the IMTP. Given that in the present study the purpose was to evaluate the efficacy of the DSI specific to lower limb dominant exercise, the IMTP was considered to be more appropriate. Furthermore, a

557 systematic reduction in DSI that would be expected if an isometric squat was employed would
558 have little influence on the key findings.

559 DSI calculated using peak force in the CMJ is commonplace (2,31,48) likely given that
560 the CMJ reflects the contribution of the stretch shortening cycle which is important to lower
561 limb power activity. However, there may be value in determining the associations considered
562 in the present study using the SJ. Although not different to CMJ determined DSI, Comfort, et
563 al. (12) demonstrated that the reliability of SJ derived DSI was marginally reduced in youth,
564 attributed to an inability to control a countermovement when executing the SJ. Novel DSI
565 calculations have not been considered using SJ derived DSI and could in fact be more
566 informative than CMJ derived DSI. Specifically, a SJ derived force-time epoch DSI would be
567 substantially lower than that of the CMJ equivalent, where in the case of the latter, during the
568 concentric phase of the movement the muscle is already highly active in the prior braking
569 phase. The theoretical constructs of the DSI are focused on evaluating the ability to express
570 maximal force producing capacity in a ballistic task where the ability to generate force rapidly
571 should be integral to this. Given the SJ measures the ability to rapidly generate propulsive
572 force from a minimal baseline in a similar manner to the IMTP, epoch focused DSI using the
573 SJ may be more strongly associated with lower limb power focused performance outcomes.
574 Furthermore, peak force and temporal force-time characteristics determined in the IMTP occur
575 at different joint angles to matched outcomes determined in the vertical jump assessment,
576 where peak force occurs close to the point of maximal flexion. Examining the association
577 between DSI calculations from an isometric and vertical jump task performed at corresponding
578 joint angles, as per the approach adapted by Pleša, et al. (37), may represent a more
579 ecologically valid approach.

580 Although the clustering approach adopted in the study is a distinct strength,
581 participants were only clustered by peak CMJ and IMTP net force across the bilateral and
582 unilateral assessments and not each of the temporal based DSI calculations. Whilst this
583 approach may result in the formation of several new groups, the approach adopted was
584 considered appropriate to stratify participants into low and high strength and power groups.
585 However, whilst this clustering approach is considered appropriate to understand the efficacy
586 of the DSI in participants that differ in the underlining strength and power characteristics, the
587 findings are specific to recreationally active, non-specifically resistance trained males and may
588 not directly extrapolate to professional athletes. Furthermore, though well-established
589 approaches exist to quantify performance in the sprint and CoD task, one particular limitation
590 of using timing gates is an inability to quantify the characteristics of the start of the movement
591 prior to the position of the first timing gate. This movement time is subsequently not considered

592 in the measured performance outcome. Several constructs of the DSI calculations may be
593 specifically relevant to this phase and should be the context of future investigation.

594 *Practical Applications*

595 Results of the present study are useful to inform practitioners regarding the application
596 of DSI for athlete profiling, monitoring and exercise prescription. Practitioners should be aware
597 that several variants of DSI calculations exist, many of which are distinct and reflective of
598 different force producing constructs, and should not be considered in line with the threshold
599 variables for exercise prescription previously proposed given that several temporal DSI
600 calculations result is DSI outcomes >1.0 (44). DSI_T elicits acceptable between session
601 reliability, whilst the reliability of temporal based DSI calculations is poor to moderate,
602 indicating only the former may be suitable for long term athlete monitoring. Irrespective of
603 calculation method, when participants are grouped by root metrics DSI appears to have little
604 association with sprint and CoD performance. Of the outcomes assessed, DSI_{100} measured
605 on the dominant leg had the strongest association with the performance outcomes, which is
606 likely specific to participants that express a high CMJ propulsive peak net force. Collectively,
607 the results infer that practitioners should carefully consider the value of DSI as an exercise
608 prescription tool given limited association to sports specific power-based tasks.

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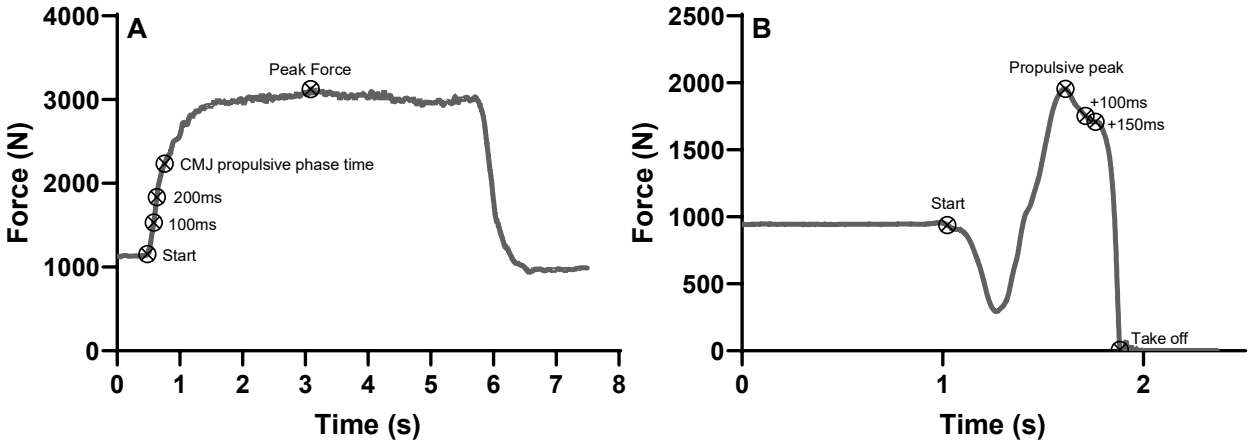
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740 **Figures**

741 Figure 1. Example of IMTP (A) and CMJ (B) force-time curves and key events determined for
742 the calculation of required outcome variables.



743

744 Figure 1.

Tables

Table 1. Number of observations and clusters identified for *K*-means cluster analysis

	High Group	Low Group
Bilateral CMJ Peak Force: No. of observations in each cluster	14	24
Bilateral CMJ Peak Force: K means cluster centers (au)	1.07	-0.63
Bilateral IMTP Peak Force: No. of observations in each cluster	19	19
Bilateral IMTP Peak Force: K means cluster centers (au)	0.86	-0.86
Dominant CMJ Peak Force: No. of observations in each cluster	9	29
Dominant CMJ Peak Force: K means cluster centers (au)	1.50	-0.46
Dominant IMTP Peak Force: No. of observations in each cluster	15	23
Dominant IMTP Peak Force: K means cluster centers (au)	1.02	-0.67
Non-Dominant CMJ Peak Force: No. of observations in each cluster	14	24
Non-Dominant CMJ Peak Force: K means cluster centers (au)	1.03	-0.60
Non-Dominant CMJ Peak Force: No of observations in each cluster	13	25
Non-Dominant CMJ Peak Force: K means cluster centers (au)	1.16	-0.59

Table 2. Test re-test reliability of bilateral and unilateral CMJ force-time metrics

		Mean ± SD		ICC			Bland-Altman		
		Session 1	Session 2	ICC	95% HDI		Bias	LoA (95% HDI)	
					Lower	Upper		Lower	Upper
Bilateral	Peak Net Force	866.03 ± 164.59	883.89 ± 171.49	0.80	0.66	0.89	17.74	-210.35	206.80
	100ms Mean Net Force	755.69 ± 173.63	770.55 ± 169.40	0.74	0.55	0.87	14.95	-192.77	225.97
	150ms Mean Net Force	722.16 ± 167.55	734.84 ± 173.60	0.77	0.59	0.88	12.42	-205.46	233.16
	Propulsive Mean Net Force	578.33 ± 119.00	577.01 ± 129.93	0.78	0.61	0.88	-1.38	-156.37	156.10
	Propulsive Net Impulse	168.81 ± 20.46	163.93 ± 20.08	0.64	0.41	0.81	-4.88	-39.92	27.38
Non-Dominant	Peak Net Force	452.12 ± 121.91	435.85 ± 98.56	0.57	0.32	0.77	16.27	-211.93	182.12
	100ms Mean Net Force	354.34 ± 117.84	347.69 ± 117.34	0.68	0.46	0.83	-6.49	-189.23	178.11
	150ms Mean Net Force	348.28 ± 114.64	342.38 ± 115.05	0.67	0.45	0.82	-5.82	-186.97	170.82
	Propulsive Mean Net Force	251.77 ± 82.53	238.85 ± 70.92	0.62	0.29	0.81	-12.89	-134.41	110.74
	Propulsive Net Impulse	88.35 ± 25.50	80.39 ± 18.18	0.50	0.22	0.71	-7.972	-50.7	35.62
Dominant	Peak Net Force	432.74 ± 107.62	439.98 ± 109.17	0.66	0.46	0.81	7.42	-163.64	179.82
	100ms Mean Net Force	359.91 ± 105.62	361.52 ± 118.15	0.67	0.45	0.82	1.85	-174.89	178.42
	150ms Mean Net Force	355.49 ± 100.15	357.06 ± 114.01	0.64	0.39	0.81	1.65	-177.85	177.57
	Propulsive Mean Net Force	244.62 ± 71.14	244.79 ± 76.67	0.72	0.52	0.86	0.24	-101.16	102.77
	Propulsive Net Impulse	81.91 ± 24.98	81.26 ± 20.24	0.30	0.01	0.58	-0.62	-51.33	50.62

[Data presented as Mean±SD; N=38] ICC = Interclass Correlation Coefficient; LOA = Limits of Agreement; HDI = Higher Density Intervals

Table 3. Test re-test reliability of bilateral and unilateral IMTP force-time metrics

		Mean ± SD		ICC			Bland-Altman		
		Session 1	Session 2	ICC	95% HDI		Bias	LoA (95% HDI)	
					Lower	Upper		Lower	Upper
Bilateral	Peak Net Force	1728.20 ± 431.90	1724.91 ± 457.17	0.82	0.69	0.90	-3.46	-523.34	529.587
	100ms Mean Net Force	288.47 ± 115.95	289.38 ± 106.23	0.48	0.18	0.70	0.60	-215.35	218.43
	150ms Mean Net Force	387.71 ± 152.85	383.57 ± 142.96	0.52	0.23	0.73	-4.01	-357.40	323.62
	Mean Net Force	646.44 ± 198.99	627.75 ± 191.82	0.61	0.39	0.79	-18.82	-363.49	311.81
	Net Impulse	195.02 ± 68.44	185.49 ± 75.83	0.78	0.63	0.88	-3.68	-92.26	96.30
Non-Dominant	Peak Net Force	1367.76 ± 377.63	1384.95 ± 349.61	0.80	0.64	0.90	17.30	-428.33	451.27
	100ms Mean Net Force	251.78 ± 84.07	249.77 ± 77.69	0.31	0.02	0.58	-2.23	-188.78	179.39
	150ms Mean Net Force	322.93 ± 107.31	323.88 ± 102.45	0.33	0.02	0.60	0.94	-229.71	232.032
	Mean Net Force	579.21 ± 167.16	577.04 ± 193.18	0.37	0.01	0.63	-1.87	-396.97	387.76
	Net Impulse	210.50 ± 76.28	209.90 ± 105.16	0.47	0.18	0.69	-1.94	-397.46	402.41
Dominant	Peak Net Force	1400.11 ± 315.30	1412.62 ± 375.53	0.86	0.75	0.93	12.60	-336.83	362.13
	100ms Mean Net Force	263.42 ± 72.40	261.51 ± 92.27	0.29	0.01	0.57	-1.94	-192.34	190.57
	150ms Mean Net Force	349.43 ± 100.06	341.45 ± 120.25	0.40	0.06	0.65	-8.13	-263.92	244.93
	Mean Net Force	606.57 ± 154.32	578.22 ± 184.33	0.39	0.09	0.65	-28.63	-396.95	349.41
	Net Impulse	218.88 ± 85.80	208.32 ± 106.75	0.46	0.13	0.71	-10.70	201.69	174.37

[Data presented as Mean±SD; N=38] ICC = Interclass Correlation Coefficient; LOA = Limits of Agreement; HDI = Higher Density Intervals

Table 4. Test re-test reliability values for bilateral and Unilateral DSI

		Mean ± SD		ICC			Bland-Altman		
		Session 1	Session 2	ICC	95% HDI		Bias	LoA (95% HDI)	
					Lower	Upper		Lower	Upper
Bilateral	DSI_T	0.53 ± 0.16	0.54 ± 0.15	0.72	0.53	0.85	0.01	-0.22	0.23
	DSI_{100}	2.96 ± 1.17	2.98 ± 1.23	0.51	0.22	0.72	0.03	-2.23	2.33
	DSI_{150}	2.11 ± 0.89	2.20 ± 1.06	0.50	0.22	0.72	0.10	-1.79	2.03
	DSI_P	1.01 ± 0.51	1.07 ± 0.65	0.67	0.46	0.82	0.07	-0.85	0.98
	DSI_I	1.01 ± 0.51	1.09 ± 0.65	0.64	0.44	0.82	0.08	-0.87	1.02
Non-Dominant	DSI_T	0.36 ± 0.15	0.34 ± 0.13	0.76	0.59	0.87	-0.02	-0.22	0.17
	DSI_{100}	1.56 ± 0.75	1.64 ± 1.11	0.35	0.03	0.62	0.08	-1.98	2.18
	DSI_{150}	1.21 ± 0.62	1.24 ± 0.77	0.36	0.04	0.62	0.03	-1.53	1.57
	DSI_P	0.49 ± 0.27	0.51 ± 0.36	0.33	0.02	0.61	0.02	-0.70	0.79
	DSI_I	0.50 ± 0.27	0.51 ± 0.36	0.33	0.03	0.60	0.02	-0.70	0.76
Dominant	DSI_T	0.32 ± 0.11	0.33 ± 0.11	0.73	0.55	0.86	0.00	-0.15	0.15
	DSI_{100}	1.48 ± 0.64	1.61 ± 0.91	0.24	<0.01	0.53	0.13	-1.68	2.01
	DSI_{150}	1.12 ± 0.52	1.22 ± 0.68	0.29	<0.01	0.57	0.10	-1.29	1.49
	DSI_P	0.44 ± 0.20	0.50 ± 0.31	0.52	0.25	0.73	0.06	-0.44	0.57
	DSI_I	0.44 ± 0.22	0.51 ± 0.31	0.53	0.26	0.74	0.07	-0.44	0.58

[Data presented as Mean±SD; N=38] ICC = Interclass Correlation Coefficient; LOA = Limits of Agreement; HDI = Higher Density Intervals

Table 5. Test re-test reliability for sprints and COD assessments

		Mean ± SD		ICC			Bland-Altman		
		Session 1	Session 2	ICC	95% HDI		Bias	LoA (95%HDI)	
					Lower	Upper		Lower	Upper
Sprints / COD	5M	1.08 ± 0.07	1.09 ± 0.08	0.61	0.36	0.80	0.01	-0.12	0.13
	10M	1.84 ± 0.09	1.86 ± 0.11	0.69	0.44	0.84	0.01	-0.15	0.16
	20M	3.23 ± 0.18	3.23 ± 0.18	0.81	0.66	0.90	-0.01	-0.22	0.16
	5-0-5 Agility	2.53 ± 0.17	2.49 ± 0.16	0.75	0.56	0.87	-0.03	-0.25	0.19

[Data presented as Mean±SD; N=38] ICC = Interclass Correlation Coefficient; LOA = Limits of Agreement; HDI = Higher Density Intervals

Table 6. Regression analysis between bilateral DSI calculations and sprint / COD data for clusters determined by IMTP peak net force

		R^2 (95% HDI)	Group Slopes (95% HDI of a Difference)	Slope Group Probability of a Difference (%)
5-0-5	DSI_T	0.11 (<0.01-0.28)	**H**: -0.12 ± 0.81; **L**: 0.08 ± 0.82 (-1.37-0.49)	**H**: 0.58; **L**: 0.42
	DSI_{100}	0.08 (<0.01-0.22)	**H**: -0.34 ± 0.12; **L**: 0.51 ± 1.07 (-2.66-2.5)	**H**: 0.64; **L**: 0.36
	DSI_{150}	0.06 (<0.01-0.16)	**H**: -0.17 ± 1.02; **L**: -0.32 ± 0.90 (-2.20-2.35)	**H**: 0.37; **L**: 0.63
	DSI_P	0.05 (<0.01-0.14)	**H**: -0.23 ± 0.79; **L**: -0.24 ± 0.60 (-1.83-1.49)	**H**: 0.43; **L**: 0.57
	DSI_I	0.05 (0.03-0.15)	**H**: -0.24 ± 0.78; **L**: -0.35 ± 0.64 (-1.61-1.69)	**H**: 0.38; **L**: 0.62
5M	DSI_T	0.11 (0.07-0.29)	**H**: -0.12 ± 0.81; **L**: 0.09 ± 0.82 (-1.32-0.49)	**H**: 0.63; **L**: 0.37
	DSI_{100}	0.12 (0.01-0.29)	**H**: 0.00 ± 1.33; **L**: 0.02 ± 1.35 (-3.54-2.23)	**H**: 0.59; **L**: 0.41
	DSI_{150}	0.07 (<0.01-0.21)	**H**: 0.04 ± 1.24; **L**: 0.32 ± 1.23 (-3.00-2.26)	**H**: 0.59; **L**: 0.41
	DSI_P	0.06 (<0.01-0.18)	**H**: -0.02 ± 1.07; **L**: 0.10 ± 1.07 (-2.60-1.48)	**H**: 0.53; **L**: 0.47
	DSI_I	0.06 (<0.01-0.19)	**H**: 0.01 ± 1.02; **L**: 0.06 ± 1.02 (-2.29-1.61)	**H**: 0.55; **L**: 0.45
10M	DSI_T	0.11 (0.01-0.28)	**H**: -0.15 ± 0.48; **L**: -0.01 ± 0.36 (-1.16-0.45)	**H**: 0.58; **L**: 0.42
	DSI_{100}	0.10 (0.01-0.25)	**H**: 0.09 ± 1.22; **L**: 0.04 ± 1.15 (-3.17-2.22)	**H**: 0.40; **L**: 0.60
	DSI_{150}	0.07 (<0.01-0.19)	**H**: 0.10 ± 1.09; **L**: 0.05 ± 1.00 (-2.67-2.18)	**H**: 0.40; **L**: 0.60
	DSI_P	0.05 (<0.01-0.17)	**H**: -0.05 ± 0.84; **L**: -0.04 ± 0.72 (-1.92-1.41)	**H**: 0.57; **L**: 0.43
	DSI_I	0.06 (<0.01-0.17)	**H**: -0.02 ± 0.82; **L**: -0.09 ± 0.70 (-1.70-1.66)	**H**: 0.40; **L**: 0.60
20M	DSI_T	0.11 (0.01-0.28)	**H**: -0.32 ± 0.51; **L**: -0.16 ± 0.35 (-1.20-0.47)	**H**: 0.60; **L**: 0.40
	DSI_{100}	0.10 (0.01-0.24)	**H**: -0.01 ± 1.11; **L**: -0.09 ± 1.05 (-2.86-2.25)	**H**: 0.40; **L**: 0.60
	DSI_{150}	0.06 (<0.01-0.18)	**H**: -0.04 ± 1.05; **L**: -0.10 ± 0.95 (-2.49-2.15)	**H**: 0.40; **L**: 0.60
	DSI_P	0.05 (<0.01-0.16)	**H**: -0.18 ± 0.82; **L**: -0.17 ± 0.68 (-1.96-1.46)	**H**: 0.58; **L**: 0.42
	DSI_I	0.05 (<0.0010.16)	**H**: -0.19 ± 0.82; **L**: -0.27 ± 0.69 (-1.70-1.81)	**H**: 0.39; **L**: 0.61

L = low bilateral IMTP peak net force; H = high bilateral IMTP peak net force.
An increasingly negative group slope indicates a higher probability of reduced time to complete the task when compared to higher slope values.

Table 7. Regression analysis between unilateral DSI calculations and sprint / COD data for clusters determined by IMTP peak net force

		Non-dominant			Dominant		
		R^2 (95% HDI)	Group Slopes (95% HDI of a Difference)	Slope Group Probability of a Difference (%)	R^2 (95% HDI)	Group Slopes (95% HDI of a Difference)	Slope Group Probability of a Difference (%)
5-0-5	DSI_T	0.21 (0.05-0.45)	**H**: -0.16 ± 0.19; **L**: -0.11 ± 0.26 (-0.69-0.45)	**H**: 0.54; **L**: 0.46	0.26 (0.07-0.45)	**H**: -0.11 ± 0.19; **L**: -0.04 ± 0.20 (-0.69-0.37)	**H**: 0.55; **L**: 0.45
	DSI_{100}	0.05 (<0.01-0.18)	**H**: -0.19 ± 0.78; **L**: -0.11 ± 0.84 (-2.32-1.59)	**H**: 0.50; **L**: 0.50	0.10 (<0.01-0.27)	**H**: -1.58 ± 1.51; **L**: -1.46 ± 1.43 (-3.01-2.23)	**H**: 0.50; **L**: 0.50
	DSI_{150}	0.05 (<0.01-0.18)	**H**: -0.20 ± 0.74; **L**: -0.11 ± 0.83 (-2.31-1.60)	**H**: 0.51; **L**: 0.49	0.06 (<0.01-0.19)	**H**: -0.72 ± 0.98; **L**: -0.72 ± 0.97 (-2.10-1.93)	**H**: 0.46; **L**: 0.54
	DSI_P	0.04 (<0.01-0.13)	**H**: -0.16 ± 0.74; **L**: -0.05 ± 0.42 (-1.29-0.73)	**H**: 0.56; **L**: 0.44	0.04 (<0.01-0.13)	**H**: -0.16 ± 0.33; **L**: -0.05 ± 0.42 (-1.30-0.73)	**H**: 0.56; **L**: 0.44
	DSI_I	0.03 (<0.01-0.13)	**H**: -0.14 ± 0.33; **L**: -0.04 ± 0.42 (-1.28-0.71)	**H**: 0.56; **L**: 0.44	0.04 (<0.01-0.13)	**H**: -0.14 ± 0.33; **L**: -0.04 ± 0.42 (-1.27-0.71)	**H**: 0.56; **L**: 0.44
5M	DSI_T	0.20 (0.01-0.43)	**H**: -0.07 ± 0.72; **L**: 0.04 ± 0.73 (-0.83-0.37)	**H**: 0.59; **L**: 0.41	0.20 (0.05-0.43)	**H**: -0.04 ± 0.72; **L**: 0.01 ± 0.73 (-0.74-0.50)	**H**: 0.52; **L**: 0.48
	DSI_{100}	0.19 (0.03-0.37)	**H**: -0.05 ± 1.07; **L**: 0.02 ± 1.07 (-2.23-1.75)	**H**: 0.58; L: 0.42	0.07 (<0.01-0.25)	**H**: 0.02 ± 1.10; **L**: 0.09 ± 1.09 (-2.44-1.85)	**H**: 0.51; **L**: 0.49
	DSI_{150}	0.10 (<0.01-0.25)	**H**: -0.02 ± 1.00; **L**: 0.04 ± 1.00 (-2.03-1.56)	**H**: 0.50; **L**: 0.50	0.03 (<0.01-0.14)	**H**: -0.04 ± 1.05; **L**: 0.06 ± 1.06 (-2.19-1.54)	**H**: 0.53; **L**: 0.47
	DSI_P	0.03 (0.001-0.11)	**H**: 0.02 ± 0.84; **L**: 0.03 ± 0.84 (-0.96-0.91)	**H**: 0.51; **L**: 0.49	0.04 (<0.01-0.15)	**H**: -0.05 ± 0.92; **L**: 0.07 ± 1.06 (-1.53-0.93)	**H**: 0.56; **L**: 0.44
	DSI_I	0.02 (<0.01-0.08)	**H**: -0.01 ± 0.98; **L**: 0.07 ± 0.90 (-1.26-0.92)	**H**: 0.55; **L**: 0.45	0.04 (<0.01-0.15)	**H**: -0.04 ± 0.92; **L**: 0.07 ± 0.91 (-1.55-0.97)	**H**: 0.55; **L**: 0.45
10M	DSI_T	0.25 (0.07-0.44)	**H**: -0.12 ± 0.30; L: -0.02 ± 0.32 (-0.79-0.36)	**H**: 0.52; **L**: 0.48	0.20 (0.05-0.44)	**H**: -0.20 ± 0.36; **L**: -0.14 ± 0.38 (-0.75-0.44)	**H**: 0.53; **L**: 0.47
	DSI_{100}	0.13 (0.01-0.30)	**H**: -0.59 ± 1.29; L: -0.53 ± 1.19 (-2.37-1.86)	**H**: 0.52; **L**: 0.48	0.05 (<0.01-0.18)	**H**: -0.19 ± 0.93; **L**: -0.09 ± 0.97 (-2.52-1.65)	**H**: 0.51; **L**: 0.49
	DSI_{150}	0.08 (<0.01-0.21)	**H**: -0.43 ± 1.11; L: -0.42 ± 1.04 (-1.93-1.70)	**H**: 0.53; **L**: 0.47	0.03 (<0.01-0.11)	**H**: -0.18 ± 0.82; **L**: -0.05 ± 0.86 (-2.12-1.33)	**H**: 0.54; **L**: 0.46
	DSI_P	0.03 (<0.01-0.11)	**H**: -0.07 ± 0.50; L: 0.06 ± 0.57 (0.90-0.91)	**H**: 0.52; **L**: 0.48	0.03 (<0.01-0.14)	**H**: -0.23 ± 0.58; **L**: -0.10 ± 0.59 (-1.40-0.81)	**H**: 0.56; **L**: 0.44
	DSI_I	0.02 (<0.01-0.09)	**H**: 0.00 ± 0.55; L: -0.08 ± 0.60 (1.15-0.89)	**H**: 0.47; **L**: 0.53	0.03 (<0.01-0.13)	**H**: -0.23 ± 0.60; **L**: -0.08 ± 0.60 (-1.57-0.82)	**H**: 0.57; **L**: 0.43
20M	DSI_T	0.25 (0.07-0.44)	**H**: -0.12 ± 0.25; **L**: -0.03 ± 0.25 (-0.76-0.34)	**H**: 0.57; **L**: 0.43	0.20 (0.05-0.41)	**H**: -0.23 ± 0.59; **L**: -0.08 ± 0.31 (-0.67-0.43)	**H**: 0.53; **L**: 0.47
	DSI_{100}	0.17 (0.02-0.36)	**H**: -0.36 ± 1.08; **L**: -0.46 ± 0.99 (-1.94-2.21)	**H**: 0.42; **L**: 0.58	0.08 (<0.01-0.28)	**H**: 0.03 ± 0.86; **L**: 0.06 ± 0.90 (-2.23-1.84)	**H**: 0.53; **L**: 0.47
	DSI_{150}	0.05 (<0.01-0.19)	**H**: 0.07 ± 0.72; **L**: 0.13 ± 0.81 (-1.92-1.45)	**H**: 0.50; **L**: 0.50	0.05 (<0.01-0.19)	**H**: 0.07 ± 0.72; **L**: 0.13 ± 0.81 (-1.92-1.45)	**H**: 0.50; **L**: 0.50
	DSI_P	0.05 (<0.01-0.16)	**H**: -0.11 ± 0.45; **L**: -0.01 ± 0.49 (-1.34-0.82)	**H**: 0.56; **L**: 0.44	0.04 (<0.01-0.16)	**H**: -0.11 ± 0.45; **L**: -0.01 ± 0.49 (-1.34-0.82)	**H**: 0.56; **L**: 0.44
	DSI_I	0.04 (<0.01-0.16)	**H**: -0.08 ± 0.47; **L**: 0.01 ± 0.49 (-1.34-0.52)	**H**: 0.53; **L**: 0.47	0.04 (<0.01-0.16)	**H**: -0.08 ± 0.47; **L**: 0.01 ± 0.52 (-1.43-0.85)	**H**: 0.53; **L**: 0.47

L = low bilateral IMTP peak net force; H = high bilateral IMTP peak net force.

An increasingly negative group slope indicates a higher probability of reduced time to complete the task when compared to higher slope values.

Table 8. Regression analysis between bilateral DSI calculations and sprint / COD data for clusters determined by CMJ peak net force

		R^2 (95% HDI)	Group Slopes (95% HDI of a Difference)	Slope Group Probability of a Difference (%)
5-0-5	DSI_T	0.09 (0.01-0.25)	**H**: -0.02 ± 0.22; **L**: -0.02 ± 0.24 (-0.48-0.75)	**H**: 0.48; **L**: 0.52
	DSI_{100}	0.09 (0.01-0.25)	**H**: -0.44 ± 1.08; **L**: -0.23 ± 1.07 (-2.64-2.25)	**H**: 0.63; **L**: 0.37
	DSI_{150}	0.07 (<0.01-0.20)	**H**: -0.24 ± 0.93; **L**: -0.24 ± 0.94 (-1.95-2.60)	**H**: 0.58; **L**: 0.42
	DSI_P	0.05 (<0.01-0.16)	**H**: -0.15 ± 0.67; **L**: -0.34 ± 0.67 (-1.11-2.29)	**H**: 0.48; **L**: 0.52
	DSI_I	0.05 (<0.01-0.18)	**H**: -0.25 ± 0.70; **L**: -0.41 ± 0.72 (-1.17-2.21)	**H**: 0.49; **L**: 0.51
5M	DSI_T	0.09 (<0.01-0.27)	**H**: -0.02 ± 0.78; **L**: -0.04 ± 0.79 (-0.55-0.86)	**H**: 0.48; **L**: 0.52
	DSI_{100}	0.11 (0.01-0.31)	**H**: -0.12 ± 1.16; **L**: 0.14 ± 1.15 (-2.72-2.22)	**H**: 0.65; **L**: 0.35
	DSI_{150}	0.08 (<0.01-0.24)	**H**: -0.00 ± 1.11; **L**: 0.07 ± 1.10 (-2.12-2.44)	**H**: 0.60; **L**: 0.40
	DSI_P	0.06 (<0.01-0.18)	**H**: 0.12 ± 1.02; **L**: -0.07 ± 1.00 (-1.26-2.36)	**H**: 0.48; **L**: 0.52
	DSI_I	0.06 (<0.01-0.20)	**H**: 0.12 ± 1.00; **L**: -0.04 ± 0.99 (-1.26-2.20)	**H**: 0.48; **L**: 0.52
10M	DSI_T	0.10 (0.01-0.25)	**H**: -0.08 ± 0.35; **L**: -0.17 ± 0.43 (-0.47-0.90)	**H**: 0.47; **L**: 0.56
	DSI_{100}	0.10 (0.01-0.28)	**H**: -0.11 ± 1.13; **L**: 0.16 ± 1.16 (-2.80-2.22)	**H**: 0.34; **L**: 0.67
	DSI_{150}	0.08 (<0.01-0.22)	**H**: -0.01 ± 1.03; **L**: 0.08 ± 1.03 (-2.17-2.37)	**H**: 0.60; **L**: 0.40
	DSI_P	0.05 (<0.01-0.17)	**H**: -0.09 ± 0.78; **L**: -0.27 ± 0.08 (-1.12-2.26)	**H**: 0.47; **L**: 0.53
	DSI_I	0.06 (<0.01-0.17)	**H**: -0.12 ± 0.77; **L**: -0.28 ± 0.78 (-1.17-2.25)	**H**: 0.48; **L**: 0.52
20M	DSI_T	0.11 (0.01-0.27)	**H**: -0.14 ± 0.33; **L**: -0.19 ± 0.35 (-0.49-0.75)	**H**: 0.48; **L**: 0.52
	DSI_{100}	0.11 (0.01-0.28)	**H**: -0.16 ± 1.13; **L**: 0.07 ± 1.10 (-2.54-2.29)	**H**: 0.65; **L**: 0.35
	DSI_{150}	0.11 (0.01-0.28)	**H**: -0.16 ± 1.13; **L**: 0.07 ± 1.09 (-2.54-2.28)	**H**: 0.65; **L**: 0.35
	DSI_P	0.08 (<0.01-0.23)	**H**: -0.07 ± 1.00; **L**: -0.03 ± 0.98 (-1.99-2.51)	**H**: 0.59; **L**: 0.41
	DSI_I	0.06 (<0.01-0.18)	**H**: -0.23 ± 0.82; **L**: -0.47 ± 0.80 (-1.99-2.47)	**H**: 0.45; **L**: 0.55

L = low bilateral CMJ peak net force; H = high bilateral CMJ peak net force.
An increasingly negative group slope indicates a higher probability of reduced time to complete the task when compared to higher slope values.

Table 9. Regression analysis between unilateral DSI calculations and sprint / COD data for clusters determined by CMJ peak net force

		Non-dominant			Dominant		
		R^2 (95% HDI)	Group Slopes (95% HDI of a Difference)	Slope Group Probability of a Difference (%)	R^2 (95% HDI)	Group Slopes (95% HDI of a Difference)	Slope Group Probability of a Difference (%)
5-0-5	DSI_T	0.15 (0.03-0.36)	**H**: -0.09 ± 0.22; **L**: 0.07 ± 0.30 (-1.52-0.37)	**H**: 0.73; **L**: 0.27	0.22 (0.05-0.43)	**H**: -0.02 ± 0.17; **L**: -0.03 ± 0.31 (-0.63-0.78)	**H**: 0.43; **L**: 0.57
	DSI_{100}	0.14 (0.01-0.31)	**H**: -0.02 ± 0.76; **L**: -0.12 ± 0.97 (-1.57-2.88)	**H**: 0.45; **L**: 0.55	0.34 (0.11-0.53)	**H**: -1.3 ± 1.15; **L**: -0.31 ± 1.30 (-4.62-1.26)	**H**: 0.82; **L**: 0.18
	DSI_{150}	0.10 (0.01-0.25)	**H**: 0.05 ± 0.58; **L**: 0.05 ± 0.75 (-1.48-1.92)	**H**: 0.57; **L**: 0.43	0.26 (0.05-0.48)	**H**: -0.6 ± 0.79; **L**: -0.14 ± 0.98 (-2.78-1.49)	**H**: 0.73; **L**: 0.27
	DSI_P	0.09 (0.01-0.23)	**H**: -0.13 ± 0.33; **L**: 0.14 ± 0.52 (-1.87-0.60)	**H**: 0.72; **L**: 0.28	0.13 (0.01-0.35)	**H**: -0.1 ± 0.32; **L**: -0.00 ± 0.57 (-1.26-1.24)	**H**: 0.61; **L**: 0.39
	DSI_I	0.01 (0.01-0.22)	**H**: -0.09 ± 0.34; **L**: 0.14 ± 0.53 (-1.77 -0.66)	**H**: 0.70; **L**: 0.30	0.11 (0.01-0.29)	**H**: 0.04 ± 0.37; **L**: 0.09 ± 0.64 (-1.42-1.40)	**H**: 0.59; **L**: 0.41
5M	DSI_T	0.16 (0.02-0.38)	**H**: -0.07 ± 0.87; **L**: 0.13 ± 0.86 (-1.18-0.51)	**H**: 0.73; **L**: 0.27	0.21 (0.01-0.43)	**H**: 0.01 ± 0.81; **L**: -0.03 ± 0.82 (-0.79-1.16)	**H**: 0.47; **L**: 0.53
	DSI_{100}	0.13 (0.01-0.32)	**H**: 0.13 ± 1.24; **L**: -0.10 ± 1.23 (-1.70-3.79)	**H**: 0.48; **L**: 0.52	0.36 (0.13-0.54)	**H**: -0.14 ± 1.23; **L**: 0.24 ± 1.25 (-3.04-2.03)	**H**: 0.70; **L**: 0.30
	DSI_{150}	0.09 (0.01-0.23)	**H**: 0.24 ± 1.09; **L**: 0.02 ± 1.08 (-1.79-2.26)	**H**: 0.57; **L**: 0.43	0.24 (0.01-0.44)	**H**: -0.05 ± 1.16; **L**: 0.13 ± 1.17 (-2.56 -2.37)	**H**: 0.64; **L**: 0.36
	DSI_P	0.09 (0.01-0.22)	**H**: -0.15 ± 1.03; **L**: 0.20 ± 1.03 (-2.26-0.80)	**H**: 0.73; **L**: 0.27	0.12 (0.01-0.31)	**H**: 0.04 ± 1.04; **L**: 0.04 ± 1.04 (-1.70-2.21)	**H**: 0.43; **L**: 0.57
	DSI_I	0.09 (0.01-0.22)	**H**: -0.16 ± 0.96; **L**: 0.16 ± 0.96 (-2.17-0.81)	**H**: 0.72; **L**: 0.28	0.10 (0.01-0.26)	**H**: -0.04 ± 1.03; **L**: 0.02 ± 1.05 (-1.97-2.15)	**H**: 0.60; **L**: 0.40
10M	DSI_T	0.17 (0.03-0.41)	**H**: -0.24 ± 0.42; **L**: -0.03 ± 0.44 (-1.16-0.41)	**H**: 0.75; **L**: 0.25	0.22 (0.05-0.43)	**H**: -0.06 ± 0.32; **L**: -0.07 ± 0.41 (-0.75-0.95)	**H**: 0.45; **L**: 0.55
	DSI_{100}	0.13 (0.01-0.43)	**H**: -0.06 ± 0.97; **L**: -0.17 ± 1.11 (-1.76-0.31)	**H**: 0.46; **L**: 0.54	0.33 (0.11-0.52)	**H**: -0.95 ± 1.36; **L**: -0.20 ± 1.29 (-4.42-1.56)	**H**: 0.77; **L**: 0.23
	DSI_{150}	0.08 (0.01-0.22)	**H**: -0.09 ± 0.80; **L**: -0.08 ± 0.92 (-1.64-2.05)	**H**: 0.57; **L**: 0.43	0.23 (0.01-0.43)	**H**: -0.58 ± 1.04; **L**: -0.18 ± 1.11 (-2.92-1.73)	**H**: 0.70; **L**: 0.30
	DSI_P	0.09 (0.01-0.22)	**H**: -0.36 ± 0.63; **L**: -0.01 ± 0.71 (-2.09-0.67)	**H**: 0.74; **L**: 0.26	0.11 (0.01-0.31)	**H**: -0.02 ± 0.52; **L**: 0.03 ± 0.68 (-1.47-1.56)	**H**: 0.60; **L**: 0.40
	DSI_I	0.09 (0.01-0.22)	**H**: -0.33 ± 0.66; **L**: -0.01 ± 0.72 (-2.04-0.72)	**H**: 0.74; **L**: 0.26	0.11 (0.01-0.27)	**H**: 0.20 ± 0.68; **L**: 0.16 ± 0.83 (-1.61-2.16)	**H**: 0.56; **L**: 0.44
20M	DSI_T	0.16 (0.03-0.37)	**H**: -0.18 ± 0.30; **L**: 0.02 ± 0.35 (-1.14-0.37)	**H**: 0.75; **L**: 0.25	0.23 (0.05-0.44)	**H**: -0.10 ± 0.25; **L**: -0.07 ± 0.36 (-0.81-0.81)	**H**: 0.60; **L**: 0.40
	DSI_{100}	0.15 (0.02-0.34)	**H**: 0.26 ± 0.96; **L**: 0.09 ± 1.06 (-1.65-3.21)	**H**: 0.48; **L**: 0.52	0.36 (0.11-0.52)	**H**: -0.53 ± 0.96; **L**: -0.03 ± 1.08 (-3.30-1.67)	**H**: 0.73; **L**: 0.27
	DSI_{150}	0.10 (0.01-0.25)	**H**: 0.18 ± 0.75; **L**: 0.15 ± 0.89 (-1.63-2.21)	**H**: 0.45; **L**: 0.55	0.24 (0.01-0.45)	**H**: -0.27 ± 0.76; **L**: -0.03 ± 0.92 (-2.29-1.91)	**H**: 0.67; **L**: 0.33
	DSI_P	0.09 (0.01-0.23)	**H**: -0.14 ± 0.44; **L**: -0.13 ± 0.60 (-1.83-0.67)	**H**: 0.71; **L**: 0.29	0.13 (0.01-0.33)	**H**: 0.09 ± 0.41; **L**: 0.06 ± 0.64 (-1.24-1.63)	**H**: 0.55; **L**: 0.45
	DSI_I	0.09 (0.01-0.23)	**H**: -0.11 ± 0.44; **L**: -0.15 ± 0.60 (-1.77-0.72)	**H**: 0.29; **L**: 0.71	0.12 (0.01-0.29)	**H**: 0.31 ± 0.54; **L**: 0.21 ± 0.79 (-1.50-2.12)	**H**: 0.51; **L**: 0.49

L = low bilateral CMJ peak net force; H = high bilateral CMJ peak net force.

An increasingly negative group slope indicates a higher probability of reduced time to complete the task when compared to higher slope values.

Supplementary Table1. Association between bilateral and unilateral DSI outcomes.

		Bilateral					Non-Dominant					Dominant				
		DSI_T	DSI_{100}	DSI_{150}	DSI_P	DSI_i	DSI_T	DSI_{100}	DSI_{150}	DSI_P	DSI_i	DSI_T	DSI_{100}	DSI_{150}	DSI_P	DSI_i
Bilateral	DSI_T		0.41 (0.19-0.56)	0.33 (0.12-0.50)	0.35 (0.13-0.52)	0.34 (0.12-0.51)	0.17 (0.01-0.36)	0.19 (0.02-0.38)	0.22 (0.03-0.40)	0.22 (0.03-0.40)	0.40 (0.18-0.55)	0.14 (<0.01-0.33)	0.19 (0.02-0.38)	0.22 (0.03-0.41)	0.21 (0.03-0.40)	0.23 (0.04-0.41)
	DSI_{100}			0.93 (0.92-0.94)	0.74 (0.63-0.79)	0.70 (0.57-0.76)	0.25 (0.05-0.43)	0.25 (0.05-0.43)	0.06 (<0.01-0.21)	0.33 (0.12-0.50)	0.33 (0.12-0.51)	0.26 (0.05-0.44)	0.08 (<0.01-0.25)	0.06 (<0.01-0.21)	0.33 (0.11-0.50)	0.31 (0.11-0.50)
	DSI_{150}				0.79 (0.71-0.83)	0.79 (0.70-0.83)	0.20 (0.02-0.39)	0.22 (0.04-0.41)	0.29 (0.08-0.46)	0.32 (0.11-0.49)	0.32 (0.11-0.49)	0.03 (<0.01-0.15)	0.22 (0.04-0.41)	0.28 (0.08-0.46)	0.32 (0.11-0.49)	0.32 (0.10-0.50)
	DSI_P					0.98 (0.98-0.98)	0.032 (<0.01-0.14)	0.20 (0.02-0.39)	0.32 (0.09-0.49)	0.33 (0.12-0.50)	0.41 (0.19-0.56)	0.03 (<0.01-0.15)	0.07 (<0.01-0.24)	0.07 (<0.01-0.24)	0.12 (<0.01-0.31)	0.320 (0.10-0.49)
	DSI_I						0.059 (<0.01-0.21)	0.18 (0.01-0.37)	0.28 (0.08-0.46)	0.40 (0.18-0.55)	0.38 (0.15-0.54)	0.03 (<0.01-0.13)	0.07 (<0.01-0.22)	0.07 (<0.01-0.22)	0.11 (<0.01-0.29)	0.11 (<0.01-0.29)
Non-Dominant	DSI_T							0.25 (0.05-0.44)	0.20 (0.02-0.39)	0.27 (0.07-0.45)	0.29 (0.08-0.46)	0.62 (0.45-0.71)	0.21 (0.03-0.40)	0.14 (<0.01-0.33)	0.09 (<0.01-0.27)	0.06 (<0.01-0.21)
	DSI_{100}								0.89 (0.85-0.90)	0.66 (0.52-0.74)	0.68 (0.54-0.75)	0.11 (<0.01-0.29)	0.20 (0.03-0.39)	0.17 (<0.01-0.36)	0.14 (<0.01-0.32)	0.12 (<0.01-0.29)
	DSI_{150}									0.85 (0.79-0.87)	0.84 (0.78-0.86)	0.07 (<0.01-0.24)	0.12 (<0.01-0.30)	0.11 (<0.01-0.29)	0.10 (<0.01-0.28)	0.09 (<0.01-0.26)
	DSI_P										0.84 (0.78-0.86)	0.07 (<0.01-0.23)	0.12 (<0.01-0.30)	0.11 (<0.01-0.29)	0.10 (<0.01-0.28	0.09 (<0.01-0.26)
	DSI_i											0.12 (<0.01-0.30)	0.17 (0.01-0.35)	0.15 (<0.01-0.33)	0.16 (<0.01-0.35)	0.12 (<0.01-0.30)

Dominant						
DSI_T				0.26 (0.05-0.44)	0.19 (0.02-0.38)	0.16 (<0.01-0.34)	0.09 (<0.01-0.26)
DSI_{100}					0.92 (0.89-0.92)	0.68 (0.54-0.75)	0.57 (0.38-0.67)
DSI_{150}						0.72 (0.61-0.78)	0.72 (0.60-0.78)
DSI_P							0.94 (0.92-0.94)
DSI_I							

Data represented as R^2 and numbers in brackets indicate 95% Higher Density Intervals